SoIM : AI & Machine Learning in Business Curriculum

Duration: 40 Hours

Target Audience: MBA Students

Format: Mixed (Theoretical Foundations + Hands-On Practical Implementation)

Instructor: Lead Data Scientist & GenAI Engineer

Course Overview

This 40-hour immersive course bridges the gap between complex technical AI/ML concepts and strategic business applications. Designed for future executives and business leaders, it covers everything from traditional machine learning to cutting-edge Generative AI and Agentic AI systems. Students will learn not only *how* these technologies work under the hood but *where* and *why* to apply them to drive ROI, automate workflows, and create competitive advantages.

Module 1: Demystifying AI & Machine Learning (4 Hours)

Objective: Establish a baseline understanding of AI technologies and their business impact.

Theory: History and Evolution of AI. Disambiguation: AI vs. Machine Learning vs. Deep Learning vs. Generative AI. The AI Value Chain: Hardware, Data, Infrastructure, Models, and Applications. Framework for identifying AI use cases across business verticals (Finance, Marketing, HR, Ops).

Practical/Case Study: Case Study Analysis: Amazon's recommendation engine, Netflix's personalization, and dynamic pricing models.

Workshop: "AI Opportunity Mapping" – Students brainstorm and map potential AI interventions in a traditional business workflow (e.g., retail supply chain).

Module 2: The Fuel of AI - Data Strategy & Analytics (5 Hours)

Objective: Understand the data lifecycle and how to prepare data for ML models.

Theory: The Data Lifecycle: Ingestion, Storage, Processing, and Analytics. Structured vs. Unstructured Data in the enterprise. Data Governance, Privacy (GDPR/CCPA/DPDP), and Security. Key business metrics and KPIs for data-driven organizations.

Practical : Hands-on: Introduction to Exploratory Data Analysis (EDA). Using Python to visualize data, identify trends, and handle missing values.

Module 3: Predictive Analytics - Traditional Machine Learning (7 Hours)

Objective: Learn how machines predict outcomes and discover hidden patterns.

Theory: Supervised Learning: Regression (Predicting continuous values like sales forecasting). Classification (Categorical prediction like churn or fraud detection). Unsupervised Learning: Clustering (e.g., Customer segmentation). Business metrics for ML: Accuracy, Precision, Recall, and ROI on False Positives vs. False Negatives.

Practical: Hands-on 1 (Classification): Building a Customer Churn Prediction model using Python (Scikit-Learn) or a no-code platform (e.g., H2O.ai, DataRobot).

Hands-on 2 (Clustering): Performing RFM (Recency, Frequency, Monetary) customer segmentation using K-Means clustering.

Module 4: Deep Learning & Unstructured Data Processing (4 Hours)

Objective: Understand neural networks and their applications in text and vision.

Theory: Intuition behind Artificial Neural Networks (ANN). Computer Vision (CV) – How machines understand images. Natural Language Processing (NLP) – How machines understand text. Enterprise use cases (e.g., visual quality control in manufacturing, sentiment analysis on brand perception).

Practical (2 hrs): Hands-on: Implementing a pre-trained sentiment analysis model using Hugging Face pipelines/APIs on a dataset of customer product reviews to automatically flag critical feedback.

Module 5: The Generative AI Revolution & LLMs (8 Hours)

Objective: Master the application of Large Language Models and GenAI in corporate settings.

Theory: The architecture of Transformers and foundation models (GPT, Claude, Llama). Prompt Engineering principles and advanced frameworks (Zero-shot, Few-shot, Chain-of-Thought). Introduction to Retrieval-Augmented Generation (RAG). Mitigating AI hallucinations and navigating AI Security risks.

Practical: Hands-on 1: Advanced Prompting Workshop – Automating the drafting of marketing copy, email correspondence, and financial summaries.

Hands-on 2: Building a “Chat with your Data” enterprise application. Using LangChain/LlamaIndex to create a secure RAG system that answers queries based on corporate PDF documents (e.g., HR policy manuals or SEC 10-K filings).